

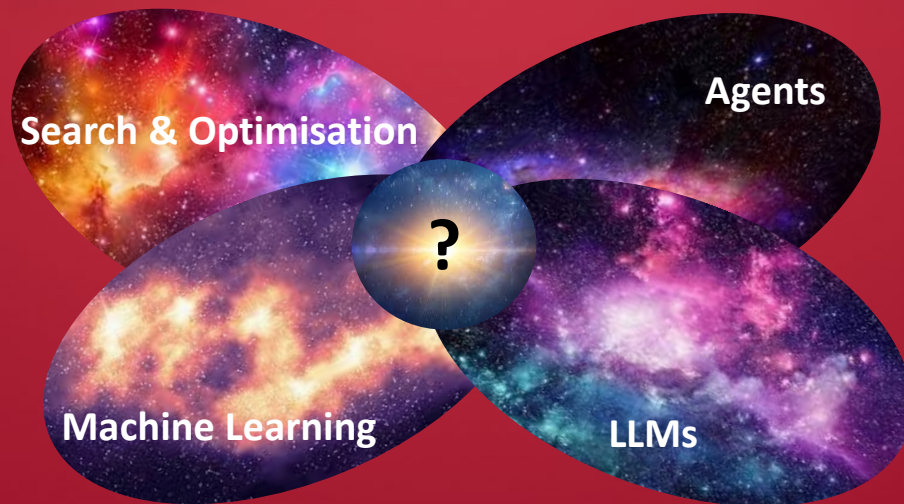
Prof. Enrique Alba

Search, optimisation, and Learning: Synergies with Agents and LLMs

Disclaimer: *this presentation does not represent in any way the positions of the affiliated organisations where the author works*

Problem and Impact

- Silos in science are bad, silos in computer science are very bad
- We need to bridge gaps and connect CS communities and knowledge



- Create a new research domain for S.O.L. automated Agentic LLMs
- Requirements:
 1. Problems need to be formulated and solved quantitatively
 2. Hierarchy of minimum two levels
 3. New methodologies for assessment of results are needed
 4. Applications should be important for society
- Results: automatic design of fine-tuned tools for complex tasks

Quantitative Framework

P8. Self-Organization of Agent Collectives

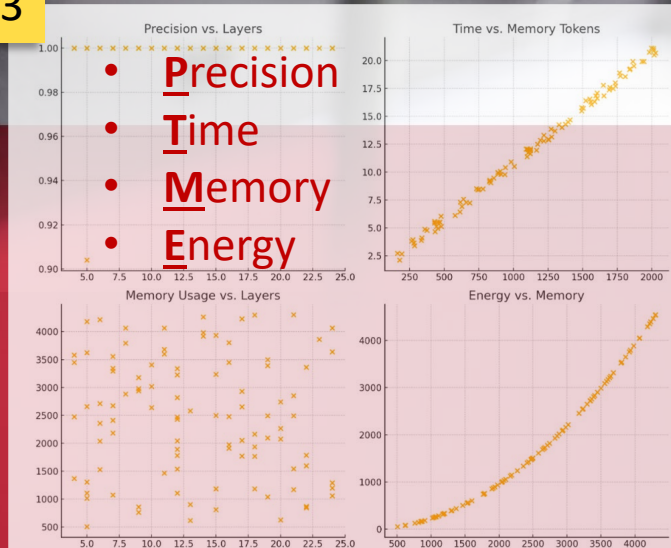
- Fitness Function:

$$f(T) = \text{TaskCompletion}(T) + \alpha \cdot \text{Scalability}(T) - \beta \cdot \text{CommunicationCost}(T)$$

- Solution Representation:

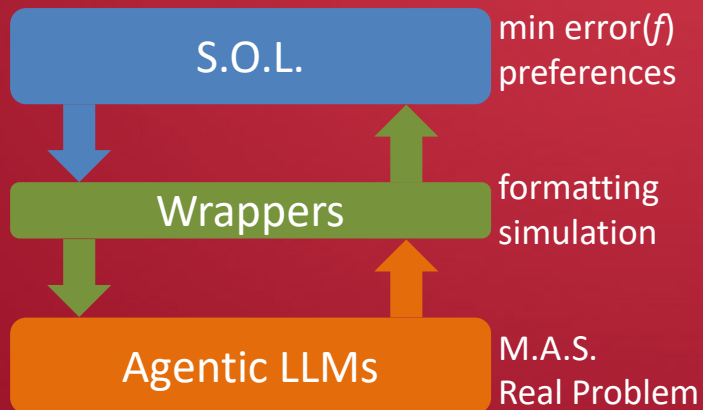
- Topology T : Adjacency matrix of interaction roles
- Type: Binary matrix or labeled graph (e.g., tree, ring, star)
- Range: $T_{i,j} \in \{0, 1\}$

R3



R1

R2



R4



Scientific
Discovery

Case Study: Scientific Discovery

8 agent scenario:

- Observation
- Question
- Hypothesis
- Experiment
- Data Collection
- Analysis
- Conclusion
- Communication

Goal min P-TME:

- Maximise precision
- Minimise time
- Minimise memory
- Minimise energy

$$\mathbf{F} = \left[-\hat{P}, T_{\text{total}}, M_{\text{total}}, E_{\text{total}} \right]$$

- $\hat{P}$: estimated **precision** of the multi-agent answer (to be **maximized** → we minimize $-\hat{P}$)
- T_{total} : total inference + communication time
- M_{total} : total memory footprint
- E_{total} : total energy usage

Multiple criteria!

$$\hat{P} = \frac{1}{N} \sum_{i=1}^N (\kappa_1 \cdot \log(L_i) + \kappa_2 \cdot \log(B_i) + \kappa_3 \cdot S_i) - \lambda \cdot \frac{m_{\text{total}}}{N}$$

Where:

- $\kappa_1, \kappa_2, \kappa_3$: empirical weights
- S_i : scientific corpus relevance factor
- λ : penalty coefficient for excessive communication

Plan C ☹ Better to simulate or run LLMs

$$T_{\text{total}} = \sum_{i=1}^N (T_i^{\text{inf}} + T_i^{\text{pre}}) + T^{\text{comm}}$$

$$T_i^{\text{inf}} = \frac{L_i \cdot B_i}{f_i} \cdot \delta, \quad T^{\text{comm}} = \tau \cdot \sum_{i=1}^N \sum_{j \neq i} m_{ij}$$

$$M_{\text{total}} = \sum_{i=1}^N (M_i^{\text{model}} + M_i^{\text{data}} + M_i^{\text{fusion}}) + M^{\text{comm}}$$

With:

- $M_i^{\text{model}} = b_1 L_i H_i^2$
- $M_i^{\text{data}} = b_2 \cdot \text{Tokens}_i$
- $M_i^{\text{fusion}} = b_3 \cdot \text{Heads}_i \cdot \text{RoleComplexity}_i$
- $M^{\text{comm}} = b_4 \cdot \sum_{i < j} m_{ij} \cdot s_{ij}$

$$E_{\text{total}} = \sum_{i=1}^N (P_i^{\text{inf}} \cdot T_i^{\text{inf}}) + E^{\text{comm}} + E^{\text{fusion}}$$

$$P_i^{\text{inf}} = \rho \cdot L_i \cdot H_i, \quad E^{\text{comm}} = \eta \cdot \sum_{i,j} m_{ij}, \quad E^{\text{fusion}} = \theta \cdot R$$

Solution Components

$$\mathbf{x} = [\text{Agent}_1, \text{Agent}_2, \dots, \text{Agent}_8, \text{CommMatrix}, R, \alpha, \gamma, \beta, \rho, \eta, \theta, \kappa_1, \kappa_2, \kappa_3, \lambda]$$

Parameter	Type	Effect of Higher Values	Effect of Lower Values
temperature	Float [0.0~2.0]	Increases randomness, creativity, and lexical diversity	Makes output more deterministic, repetitive, and factually precise
top_p	Float [0.0~1.0]	Allows more diverse tokens to be considered during sampling	Restricts generation to high-probability tokens, reducing creativity
top_k	Integer [1~100]	Expands the number of token options for sampling, increasing variability	Reduces token sampling scope, leading to safer but less varied completions
max_tokens	Integer [1~32,000]	Enables longer completions, useful for extended reasoning or dialogue	Limits generation length, reduces time and memory usage
frequency_penalty	Float [0.0~2.0]	Discourages repetition of frequent tokens, increases novelty	Allows more repetition, may improve consistency or focus
presence_penalty	Float [0.0~2.0]	Promotes introduction of new ideas or topics	Makes model stick to known or earlier content, increases conservatism
repetition_penalty	Float [1.0~2.0]	Penalizes token loops and redundant phrases	Tolerates repetition, which can help reinforce structure or coherence
stop_sequences	List[str]	Enables early stopping based on more sequences, constrains generation	Fewer stopping points, longer and potentially less focused outputs
streaming	Boolean	Enables real-time output of tokens, improves perceived responsiveness	Waits for full generation before output, increases latency
prompt_length	Integer (s context)	Improves context usage and long-term dependencies	Limits awareness of prior text, faster but less informed responses
batch_size	Integer	Increases parallelism, improves GPU throughput	Reduces throughput, may cause underutilization of compute resources
dtype	Categorical (fp16, int8, etc.)	Reduces precision and memory footprint, accelerates inference	Increases precision and accuracy, requires more memory and energy
device	Categorical (CPU, GPU)	Uses GPUs for acceleration, better performance and scalability	Uses CPUs for cost-efficiency or fallback scenarios, slower execution
use_kv_cache	Boolean	Enables token reuse for faster autoregressive decoding	Forces full recomputation at each step, slower performance
context_window	Integer (model-specific)	Allows more context to be retained, improves long-form coherence	Limits contextual information, may degrade performance on complex prompts

AgentParams

$$T(N) = N \cdot 15 + \frac{N(N-1)}{2} + 12$$

Solution Vector

$$T(8) = 8 \cdot 15 + \frac{8 \cdot 7}{2} + 12 = 120 + 28 + 12 = \boxed{160 \text{ parameters}}$$

CommMatrix

$$\text{CommMatrix} = [\mathbf{m}_{12}, \mathbf{m}_{13}, \mathbf{m}_{14}, \dots, \mathbf{m}_{78}] \quad \# 28 \text{ float values}$$

Meaning	Interpretation
Bandwidth	How much data flows between Agent i and Agent j
Latency weight	Communication delay between the two agents
Interaction frequency	How often agents collaborate during task execution
Redundancy factor	Degree of replicated effort due to communication

SystemParams

Symbol	Name	Type	Range / Example	Description / Role
$\mathbf{R}$	Resource model	Integer enum	(0: shared, 1: semi-shared, 2: dedicated)	Defines how compute resources are allocated across agents (affects latency and energy)
α (alpha)	Parallelism strategy	Integer enum	(0: model-parallel, 1: data-parallel, 2: hybrid)	Defines how agents are distributed over hardware
γ (gamma)	Scheduler policy	Integer enum	(0: FIFO, 1: RoundRobin, 2: priority-based)	Determines order of execution of agent tasks
β (beta)	LLM backend type	Integer enum	(0: GPT, 1: Claude, 2: LLaMA)	Affects base performance and token processing efficiency
ρ (rho)	Network bandwidth	Float	[1.0, 100.0] (in Gbps)	Governs inter-agent data flow and communication latency
η (eta)	Thread count per agent	Integer	[1, 32]	Controls concurrency inside each agent process
θ (theta)	Prompt pre-processing strategy	Integer enum	(0: none, 1: compressed, 2: embedded)	Affects memory use and processing cost
κ_1	KV cache policy	Integer enum	(0: disabled, 1: partial, 2: full)	Controls memory-time tradeoff in LLM autoregressive inference
κ_2	Precision format	Integer enum	(0: float32, 1: float16, 2: int8)	Reduces memory and energy at expense of precision
κ_3	Load balancing scheme	Integer enum	(0: static, 1: dynamic, 2: adaptive)	Affects task migration and system throughput
λ (lambda)	Redundancy threshold	Float	[0.0, 1.0]	Threshold for duplicated work or replicated communication (tradeoff accuracy vs. time)

System Architecture

S.O.L. tech : *NSGA-II*

Input : *problem definition [prefs]*

Output : *LLM-MAS-SD best design*

Run:

Create initial pop of solutions

Iterate on them

Compute best PTME Pareto Front

After end:

[Interactive decision making]

[Learning from all the process]

To/from quality assessment
Wrapper

Estimate total costs
-- academic

For each agent i , compute:

• Model cost:

- Time: $T_i = L_i \cdot \log(H_i) \cdot B_i$

- Memory: $M_i = \alpha \cdot L_i \cdot H_i + \gamma \cdot D_i$

- Energy: $E_i = \rho \cdot T_i \cdot M_i$

- Precision: $P_i = \kappa_1 \cdot L_i + \kappa_2 \cdot B_i + \kappa_3 \cdot S_i$

Aggregate over all agents:

$$P_{\text{total}} = \frac{1}{N} \sum_{i=1}^N P_i, \quad T_{\text{total}} = \sum_i T_i, \quad M_{\text{total}} = \sum_i M_i, \quad E_{\text{total}}^{\text{total}} = \sum_i E_i$$

Simulate M.AS. -- towards real systems

Simulator	Keyword 1	Keyword 2
Vidur	Token-level profiling	Cost modeling
LLMPerfSim [†]	Throughput	Latency modeling
FastServe	Serving latency	Energy estimation

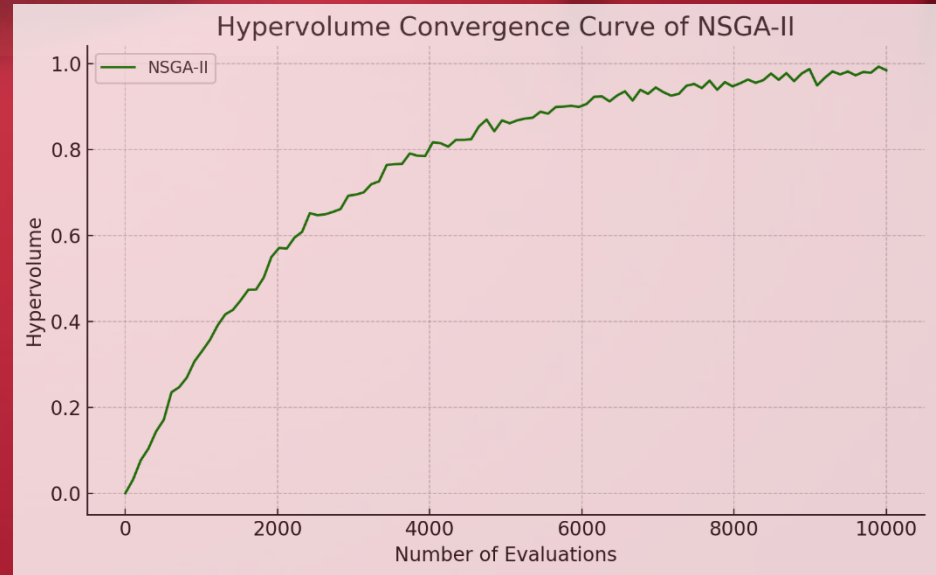
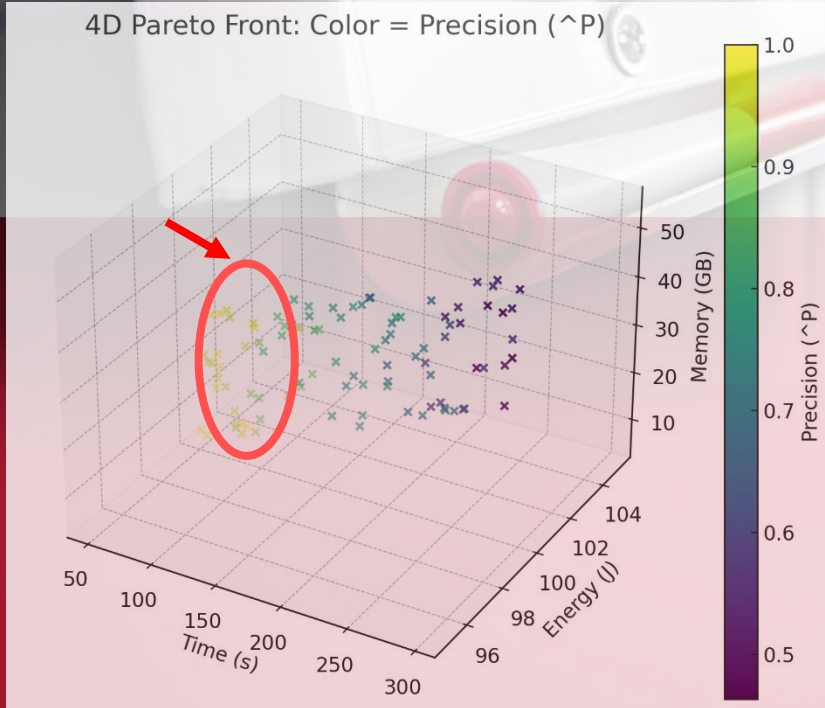
Run Agentic LLM (real)

LLaMA 3 : PyTorch, Transformers, bitsandbytes, FlashAttention

GPT-4 : OpenAI API, Python, LangChain/OpenAgents SDK

Claude 3 : Anthropic API, Python, LangChain/AutoGen

Expected Results



Warning: not real results! Just here to explain what they would look like

Limitations & Mitigations

Simulator	Supports Time	Memory	Energy	Accuracy Surrogate	Keywords
Vidur	✓	✓	✓	✗	Token-level profiling, Cost modeling
LLMPerfSim	✓	✓	✓	✗	Throughput analysis, Latency modeling
FastServe	✓	✓	✓	✗	Serving latency, Energy estimation
TokenSim	✓	✓	Partial	✗	Token ops, Transformer depth
HERMES	✓	✓	Partial	✗	Transformer latency, Streaming

L1: What is the best LLM simulator to use?

L3: S.O.L. techniques have their own hyperparameters

- sol1: theory
- sol2: literature
- sol3: autoML

L4: Scalability of the approach

- sol1: run parallel techniques at all times
- sol2: optimise the code and/or surrogate operations
- sol3: quantisation, weight patterns, MoE

L2: Simulation based profiling

- sol1: scrap real data from Internet
- sol2: used improved estimations
- sol3: mix estimations with real
- sol4: run fully real

Real Applications

Scenario 1. Autonomous Research Team for Scientific Discovery

- Field: Engineering, Science
- Pattern:  Sequential (pipeline)
- Description: LLM agents simulate literature review, hypothesis generation, experiment planning, and writing — each step done by a specialized agent.
- Costs:

$$T_{\text{total}} = \sum_i (T_i + T_{\text{handoff},i}) \quad E_{\text{total}} = \sum_i (E_i + E_{\text{handoff},i})$$

- Variables: 4 agents × 20 architectural/runtime vars + 3 handoff message sizes

Scenario 2. Co-Design Agents for Product Development

- Field: Engineering / Industrial Design
- Pattern:  Fully parallel
- Description: LLMs collaborate on design: one ideates, another critiques, others simulate or ensure constraint adherence. Shared canvas model.
- Costs:

$$T = \max_i T_i + k\delta + T_{\text{sync}} \quad E = \sum_i E_i + \sum m_{ij} \cdot (1 - \tau) \cdot \kappa$$

- Variables: 5 agents × 20 vars + 10 msg pairs + coordination cost vars

Scenario 3. Negotiation & Governance Simulation

- Field: Political Science, Urban Planning
- Pattern:  Hybrid (multi-round, async)
- Description: Agents simulate policy stakeholders negotiating energy reform or climate law.
- Costs:

$$T = R \cdot (\max_i T_i + T_{\text{msg}}^{(r)}) \quad E = R \cdot \sum_i E_i + \sum_{r=1}^R \sum_{i,j} m_{ij}^{(r)} \cdot \epsilon$$

- Variables: N agents × vars + R rounds × messaging matrices

Scenario 4. Medical Diagnosis Ensemble

- Field: Life Sciences
- Pattern:  Fully parallel
- Description: LLM agents process symptoms, patient history, genetic data, and lab results in parallel, fusing evidence.
- Costs:

$$T = \max_i T_i + T_{\text{fusion}} \quad E = \sum_i E_i + E_{\text{fusion}}$$

- Variables: 4-6 agents × vars + fusion logic + data modality sizes

Scenario 5. Historical Document Interpretation

- Field: Humanities
- Pattern:  Parallel
- Description: Agents specialize in language, script, context, and visual features of ancient manuscripts, each contributing to a joint analysis.
- Costs:

$$T = \max T_i + T_{\text{cross-check}} \quad E = \sum_i E_i + \sum \text{msg}_i$$

- Variables: 4 agents × vars + token overlaps + domain models

Scenario 6. Real-Time Policy Advising System

- Field: Political Science, Economics
- Pattern:  Parallel + Feedback Loop
- Description: During live parliamentary debate, agents generate summaries, check contradictions, propose arguments.
- Costs:

$$T = \sum_i \left(\max_j T_{ij} + \sum_j m_{ij}/\tau \right) \quad E = \text{time integral over } t$$

- Variables: Agents × vars + dialogue turns × messaging

Scenario 7. Multi-Agent Translation & Culture Engine

- Field: Humanities / NLP
- Pattern:  Hybrid
- Description: Agents work on multilingual and cultural versions of a message, each handling one culture/language and updating others in real time.
- Costs:

$$T = \max_i T_i + k \cdot \text{sync cost} \quad E = \sum_i E_i + \text{shared ontology updates}$$

- Variables: Languages × agents × translation message size

Scenario 8. Smart Infrastructure Monitoring

- Field: Civil Engineering
- Pattern:  Parallel
- Description: LLM agents analyze sensor streams from bridges, tunnels, etc., each focusing on different structure types or anomalies.
- Costs:

$$T = \max_i T_i + T_{\text{anomaly fusion}} \quad E = \sum_i E_i + E_{\text{common}}$$

- Variables: 6-10 agents × vars + signal fusion model

Scenario 9. Multi-Agent Legal Drafting System

- Field: Law / Government
- Pattern:  Sequential
- Description: Agents simulate legal drafting stages: clause generation, compliance check, ethical review, simplification.
- Costs:

$$T = \sum_i (T_i + \text{revision delay}) \quad E = \sum_i E_i + \sum \text{token exchange}$$

- Variables: 4 agents × vars + inter-clause consistency checks

Scenario 10. AI Team for Open-Ended Gameplay

- Field: Game Design / Behavioral Science
- Pattern:  Parallel + Strategic Layer
- Description: LLMs simulate autonomous NPCs with complementary roles (e.g., attacker, negotiator, builder, scout).
- Costs:

$$T = \max T_i + \text{latency}_{\text{loop}} + T_{\text{core-update}} \quad E = \sum_i E_i + \text{team coordination}$$

- Variables: N agents × behavior profile × simulation loop cost

1. Scientific Discovery Collaboration
2. Intelligent Traffic Coordination
3. Multi-LLM Judicial Analysis
4. Disaster Response Planning
5. Collaborative Medical Diagnosis
6. Cultural Heritage Preservation
7. Academic Peer Review Automation
8. Climate Policy Simulation
9. Synthetic Newsroom Generation
10. Collaborative LLM Gameplay

Agentic LLM Problems

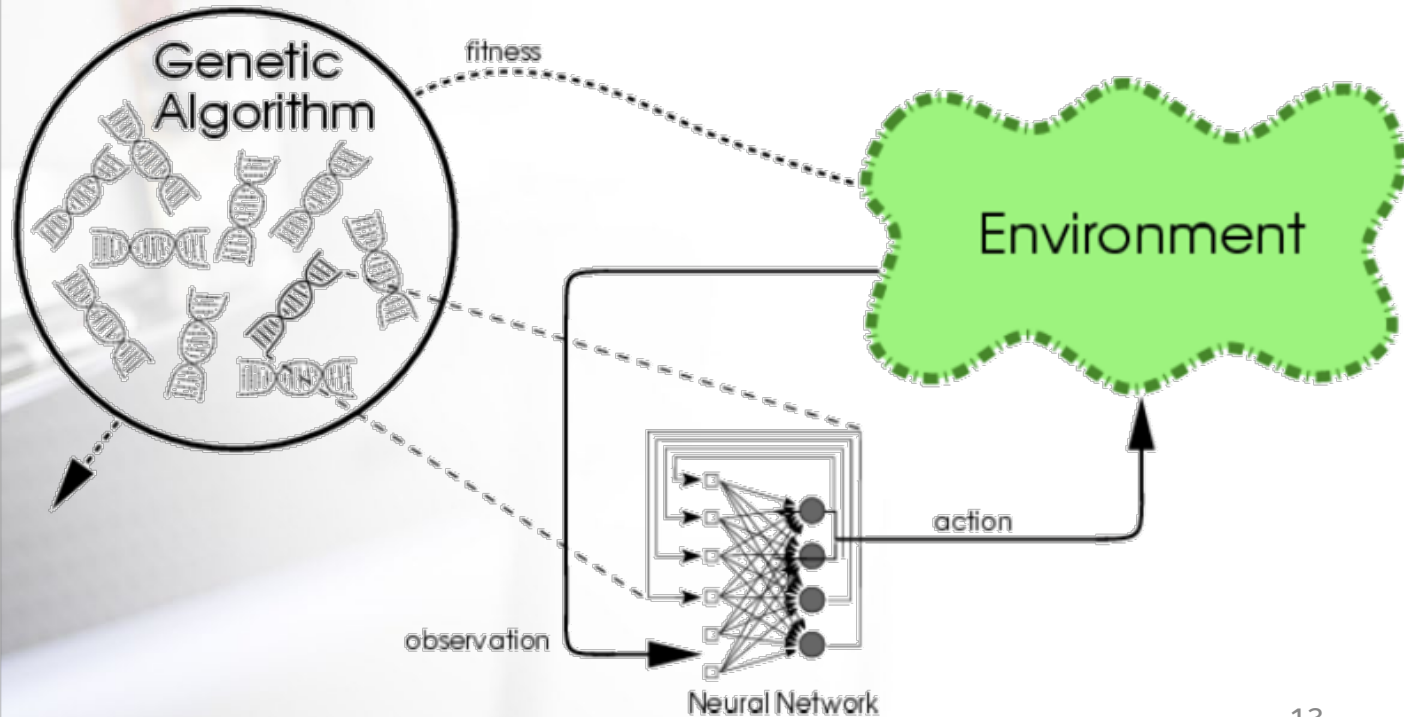
- Genetic algorithms
- Swarm intelligence
- Multiobjective
- Parallel techniques



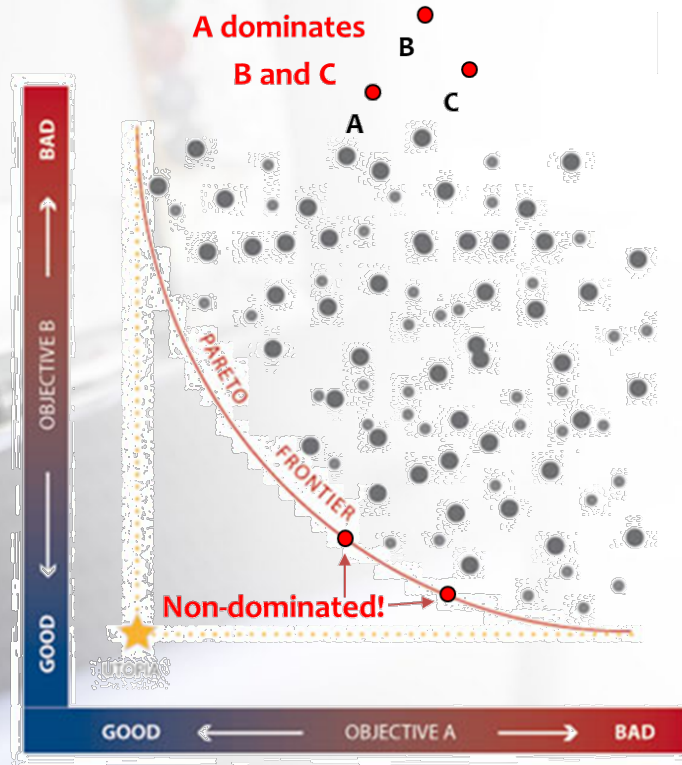
Spare Slides



Neuroevolution



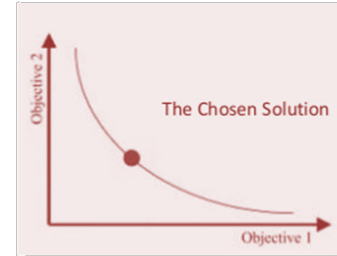
Would you Like A or B? “A and B for me, Please”



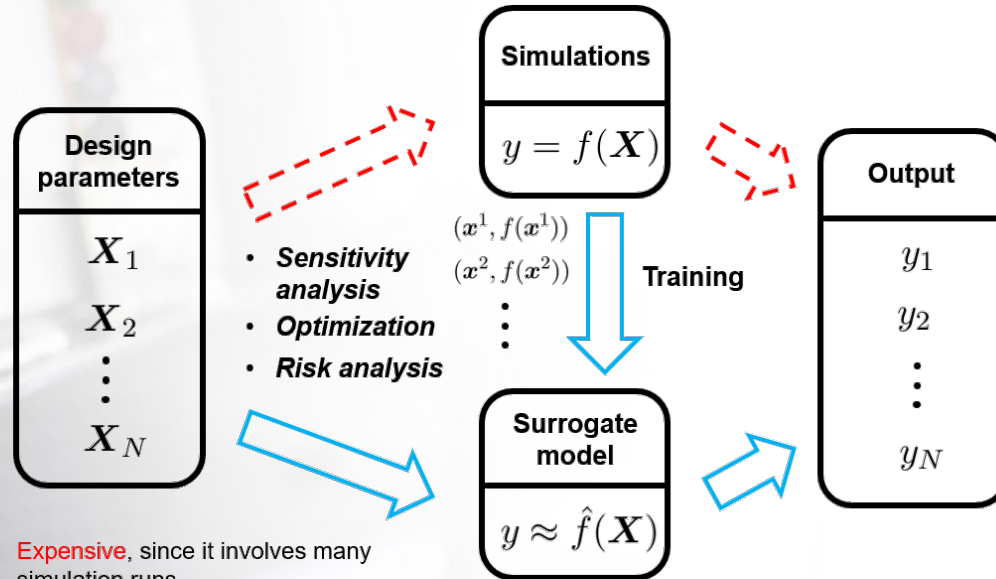
Multi-Objective Optimization:
No single “optimum” solution



Higher-level
Decision Making



Simulators and Surrogates



- **Sensitivity analysis**
- **Optimization**
- **Risk analysis**

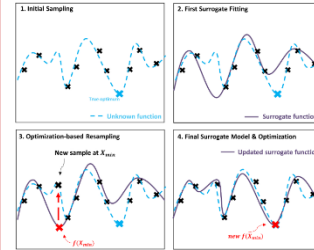


Expensive, since it involves many simulation runs

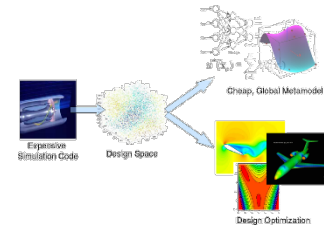


Cheap, since training and employing a surrogate model is not expensive

Fit to data...



... then use it!



Parallel Multiobjective Algorithms

- This is a **growing research line** at world level
- **Diversity preservation** is mandatory: crowding and ranking still popular
- **Distributed and M/S models** are the most popular way of building PMOEAs
- **Cellular MOEAs** are raising research line with improved spread in PFs
- **New models:** aggregated weight sums in nodes plus centralized P. Domin.
- **Programming:** C/C++, MPI, Java, MPICH-G2, MatGrid, CUDA/OpenCL

